Sapienza Università di Roma

Department of Computer Science
Master degree in Cybersecurity

ThreatLedger: Decentralized CTI Sharing on the Blockchain

Cybersecurity Seminars

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Seminars

Cyber threat intelligence: from theory to practice (Nino Vincenzo Verde, Carmine Molisso, Valerio Zottola);
Proof-of-life Through Digital Identity: The UN's Pioneering ISO 42001/27001-Certified AI Implementation (Dino Dell'Accio);
Preserving secrecy for process data exchange via public blockchain (Claudio Di Ciccio).

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1 Introduction

During my studies I developed a strong interest in blockchain technology, starting from the study of distributed systems up to its specific applications (Bitcoin, Ethereum, Smart Contracts etc.). The seminars I attended increased my curiosity, showing me how the concept of a distributed and decentralized database can offer concrete solutions to some current critical issues in the IT world.

In particular, in the seminar "*Proof-of-life Through Digital Identity: The UN's Pioneering ISO 42001/27001-Certified AI Implementation*", Professor Dino Dell'Accio discussed the use of blockchain as a distributed database, capable of guaranteeing reliability and transparency without a central authority. The Leonardo seminar, "*Cyber Threat Intelligence: from theory to practice*", made me reflect on the difficulties related to sharing intelligence information between different organizations.

Another decisive cue came from the seminar held by Professor Di Ciccio, entitled "*Preserving secrecy for process data exchange via public blockchain*", which provided me with interesting insights, showing how blockchain can facilitate the automation and traceability of complex business processes that involve multiple actors.

From these experiences, my reflection was born: How can I exploit the characteristics of blockchain to improve the sharing of CTI?
Thus, **ThreatLedger** was born, a decentralized application (DApp) designed to overcome the main critical issues in sharing CTI. The platform allows stakeholders to collaborate in a secure, transparent and anonymous way, sharing information on cyber threats in real time.

In a context where CTI plays a central role in protecting organizations from cyber attacks, but in which the management, storage, analysis and sharing of sensitive data represents a complex challenge, ThreatLedger presents itself as an innovative solution, exploiting the intrinsic properties of the blockchain.

2 Customer segments and early adopters

ThreatLedger primarily targets four major customer segments in the cybersecurity ecosystem, with an initial focus on the Italian market:

1. **Large Enterprises** operating in industries such as energy, finance, manufacturing, and telecommunications, with sophisticated IT infrastructures and large budgets dedicated to cybersecurity. These companies need effective solutions to protect themselves from cyber attacks, ensure regulatory compliance, and share intelligence securely and anonymously.

2. **Government Agencies** including defense, homeland security, and regulatory authorities, which handle sensitive data and require secure and confidential collaboration between organizations to exchange CTI.

3. **Cybersecurity Vendors** who are private companies developing cybersecurity solutions such as antivirus, SIEM, and frameworks, interested in integrating CTI to improve their products and services.

4. **Critical Infrastructure Operators** such as civil administration, healthcare, transportation, energy, and finance that provide essential services and are particularly vulnerable to cyberattacks or disruptions. They must continuously monitor emerging threats to ensure business continuity and security.

According to the Clusit 2025 Report [4], the financial, healthcare and critical infrastructure sectors in Italy were the most affected by cyber threats in 2024, underscoring the importance of educating users and adopting advanced CTI tools. As a result, ideal early adopters will be major Italian companies in the financial sector such as Intesa SanPaolo and Unicredit, public and private healthcare companies, and critical infrastructure operators such as Trenitalia, ENEL, and TIM. These customers have significant budgets to invest early in innovative solutions such as ThreatLedger, which improves the secure and timely CTI sharing, reducing risks and losses from cyberattacks. Additionally, national cybersecurity authorities such as CERT-IT or sectoral CSIRTs represent strategic stakeholders, both as trusted participants and as potential endorsers of the platform.

3 Problem statement

As defined in the NIST SP 800-150 standard [9], **Cyber Threat Intelligence (CTI) sharing** is the structured exchange of information that enables organizations to identify, assess, monitor, and respond to cyber threats. This information may include indicators of compromise (IoCs), tactics, techniques, and procedures (TTPs) employed by threat actors, recommended defensive actions, and findings derived from incident analyses. By sharing such intelligence, organizations can improve both their own security posture and that of the broader community.

In managing threat intelligence, sensitive information may be involved. Improper disclosure of such data could result in financial loss, violate laws, regulations and contracts or damage an organization's reputation. Therefore, it is essential that organizations adopt appropriate security and privacy measures, along with data management procedures to prevent unauthorized access, disclosure or modification.

The main problem is that most CTI sharing platforms are based on centralized models that introduce a **single point of failure**, put the **entire operational burden on a unique entity**, and raise serious concerns about **privacy and data control**. Since sensitive information is managed by a central authority, trust is compromised and organizations are discouraged from sharing timely and accurate threat information. Moreover, the **lack of clear incentives** for sharing makes it even more difficult to build a collaborative, functional and sustainable intelligence ecosystem. Today's organizations need a solution for secure and real-time collaboration, while ensuring confidentiality and internal security. They require a system that support automated and privacy-respecting information exchange.

The top three customer's frustrations can be summarized as follows:

1. **Fear of exposure:** Organizations fear that sharing information about attacks, discovered vulnerabilities, or incidents could result in reputational damage, legal liability, or loss of trust from customers, investors, and partners.

2. **Low trust in central authorities:** Centralized platforms require organizations to rely on a third party to manage and control sensitive data. This raises concerns about privacy, impartiality, and security. In addition, such authorities constitute a single point of failure, reducing the overall resilience of the system.

3. **Lack of incentives:** Sharing CTI is often perceived as a cost with no direct returns. There are no clear mechanisms to reward active contributors and no models to ensure fair exchange. This makes it difficult to justify engagement in sharing activities, compromising the sustainability of collaboration in the long term.

Addressing these critical issues is essential to unlocking the full potential of collaborative cybersecurity. Without a secure, transparent, and incentive-based sharing model, the CTI ecosystem will continue to be underutilized, and many organizations will be left to face increasingly complex threats on their own.

4 Existing alternatives

While many organizations still rely on informal means such as calls or emails to share CTI, there has recently been a growing interest in dedicated platforms that enable automated or semi-automated sharing of CTI data within communities. A **Threat Intelligence Platform (TIP)** is a specialized software tool designed to help organizations collect and analyze real-time threat intelligence from multiple sources, in order to support their defense strategies. In addition to TIPs, **Information Sharing and Analysis Centers (ISACs)** are organization that serve as centralized resources for collecting threat information. They can leverage TIPs and facilitate information sharing between the public and private sectors [14]. For example, in Italy, operates the National Cybersecurity Agency (ACN) [1]. Organizations can also utilize **Open Source Intelligence (OSINT)**, that is, publicly available information collected from the internet, social media, and other open channels, to enrich their threat intelligence and improve situational awareness.

Although there are numerous TIPs available on the market, most are offered under commercial licenses or rely on closed ecosystems. A number of both proprietary and open-source solutions exist, each with distinct characteristics and limitations:

Commercial Platforms

- **VirusTotal**[1] (owned by Google) is a leading CTI service that analyzes suspicious files, IP addresses, domains, and URLs to identify malware and other security threats. It generates detailed threat reports and automatically shares this information with the cybersecurity community.

- **IBM X-Force Exchange**[2] is a cloud-based collaborative CTI platform that allows organizations to access, research, and share threat intelligence including IoC, malware reports, and vulnerability information. Supported by IBM's security research team, the platform facilitates integration with existing security tools via APIs and supports open standards such as STIX and TAXII.

- **Facebook ThreatExchange**[3] is another example of a commercial CTI platform allowing selected members to share indicators and threat context. Although it facilitates collaboration among trusted partners, it requires manual onboarding and is not open to broad community contributions.

Open Source Platforms

- **MISP**[4] (Malware Information Sharing Platform) is a popular open-source platform widely used by government agencies, CERTs and the security community to collect, store, and share IoCs and CTI reports. It offers various features including a database of indicators, automated correlation, sharing capabilities, and a user-friendly interface with support for various data formats and standards. However, MISP relies heavily on trust between participants and does not provide built-in incentives or advanced anonymization mechanisms for shared information.

- **OPENCTI**[5] is another open-source platform designed to manage and visualize structured threat intelligence. It supports the full intelligence lifecycle and provides API-based integration, but, like MISP, it lacks mechanisms to ensure integrity or incentivize the sharing of high-quality data.

[1] `https://www.virustotal.com`
[2] `https://xfe-integration.xforce.ibm.com/`
[3] `https://developers.facebook.com/docs/threat-exchange/`
[4] `https://www.misp-project.org/`
[5] `https://www.opencti.io/`

- **ThreatFox**[6] is a community-driven project where security researchers and threat analysts can share IoCs, similar to MISP and VirusTotal. ThreatFox offers a feature where you can earn 5 credits for every IoC shared, up to 100 credits for a single IoC, that can be used to gain visibility within the community. These credits are provided as a form of recognition and to encourage active participation in the platform.

While many platforms and methods exist for sharing CTI, most suffer from limitations such as centralization, lack of trust mechanisms, and insufficient privacy protections. Moreover, incentive structures are not available. In this context, Liu et al [10] showed that a lack of incentives can even prevent the exchange process from happening.

Several academic studies and research papers [13, 3, 4, 11] have proposed blockchain-based models to address these issues, highlighting its potential for decentralized, transparent, and tamper-proof CTI sharing. However, to the best of my knowledge, no fully operational, widely adopted CTI sharing platforms leveraging blockchain currently exist. This gap presents a unique opportunity to explore and implement a more secure, decentralized, and incentive-driven alternative for CTI sharing.

5 Solution

ThreatLedger addresses the key challenges of sharing Cyber Threat Intelligence through a decentralized application (DApp) based on blockchain technology. This infrastructure securely records data on an immutable distributed ledger, eliminating the risks of a single point of failure and dependence on a central authority.

According to Chatziamanetoglou and Rantos [3], permissioned and private blockchains are generally preferred to ensure security and confidentiality, although permissionless models with appropriate access control mechanisms can be used. With this in mind, we chose a **permissioned consortium blockchain** model, which combines the advantages of both public and private blockchains. Only the cryptographic hash of the actual CTI is stored on-chain, while the CTI itself is kept encrypted off-chain using IPFS (InterPlanetary File System) as storage. This **hybrid approach** serves a dual purpose: it avoids overloading the blockchain with large volumes of data, improving performance and scalability; it prevents the storage of immutable sensitive information on-chain, which would raise concerns under regulations like the GDPR.

Regarding the IPFS protocol, its compliance with the GDPR, particularly the Right to be Forgotten (RTBF), remains a complex and debated topic. However, Politou et al. [12] propose an anonymous protocol for delegated content erasure

[6] `https://threatfox.abuse.ch/`

requests in IPFS, which may be included in our platform.

ThreatLedger introduces an **incentive mechanism** that avoids volatile token-based rewards or transaction fees, opting for a subscription-based system combined with discount mechanisms. This idea is inspired by recent academic research [11]; however, our approach extends it by incorporating additional incentive mechanisms such as reputation scores, priority access, exclusive analytics, and access to premium features to further encourage active and high-quality participation within the community.

Initially, the **demo version** of ThreatLedger allows potential users to explore the platform and evaluate its key features. To unlock full functionality, a Pro or Enterprise **subscription** is required, available on a monthly or annual basis. Subscribers can both submit and request CTI through transactions managed by **smart contracts**.

When a **user shares a CTI**, it receives two types of scores:

- A **quantitative score**, based on the number of transactions involving that indicator, measuring its relevance and popularity in the community;
- A **qualitative score**, determined by ratings provided by other users.

The scores are combined into **platform points**, which serve both as a reputation metric and as a reward currency within ThreatLedger.

When a **user browses a CTI** entry, initially only its metadata and cryptographic commitments are visible. If interested, the user can then request access to the full CTI content, which is stored off-chain to ensure privacy and confidentiality. Afterward, the system randomly select a limited number of users, from those who have accessed that same CTI, to evaluate its quality using predefined metrics. Only these users are rewarded with additional points, reducing the risk of opportunistic behavior aimed at accumulating points through unreliable evaluations. Notably, negative evaluations reduce the CTI score, penalizing its perceived quality and discouraging the sharing of poor-quality intelligence.

Accumulated points can be used to **obtain rewards**, such as discounts on the next subscription or access to premium features that improve the user experience. This incentive model promotes active participation, improves data quality, and helps build a collaborative and sustainable ecosystem.

To ensure successful adoption, ThreatLedger must first establish partnerships with key industry players. The team will actively participate in cybersecurity conferences and professional events to build relationships with potential consortium members. This initial phase is essential to create a strong network of trusted partners who can benefit from and contribute to the platform's ecosystem.

6 Unfair advantage

The main feature that distinguish ThreatLedger from competitors is its architecture based on **blockchain technology**, which ensures data integrity, transparency, and decentralization.

Unlike centralized solutions, that lack incentives and trust from organizations, ThreatLedger uses a permissioned consortium blockchain. This design allows organizations to retain full control over who can access sensitive information by managing decryption keys through smart contracts.

Participants can share threat data knowing it cannot be altered or accessed without explicit authorization (thanks to encryption and key management), effectively addressing privacy, legal compliance, and trust issues. The **hybrid on-chain/off-chain model** also improves the system's scalability and performance.

ThreatLedger adopts a **unique incentive system**: by sharing and rating CTI, users gain benefits such as discounts on upcoming subscriptions, reputation level, and gamification elements. This encourages active and quality participation without relying on volatile tokens. Subscriptions make using the platform simpler and more predictable compared to paying with currency for each individual transaction. Community members collaborate with confidence, knowing their data is protected and their contributions are valued.

7 Unique Value proposition

"*Share and access Cyber Threat Intelligence securely with ThreatLedger - the blockchain-powered platform that guarantees privacy, decentralization, and rewards for your contribution.*"

Listed below are all the key elements of the sentence and why they are important:

- **Share and access cyber threat intelligence securely**: Clearly describes the main functionality of the platform: a space where organizations can exchange actionable CTI. The word "securely" emphasizes data protection through encryption and access control.

- **Blockchain-powered platform that guarantees privacy, decentralization**: Highlights the innovative use of blockchain to ensure decentralization, immutability, and traceability.

- **Rewards for your contribution**: Shows that ThreatLedger goes beyond simply sharing and accessing CTI, by rewarding users with gamified incentives and subscription benefits.

This sentence encapsulates ThreatLedger's unique combination of technological power and user-centered design, explaining why customers should choose to collaborate and use our platform over competitors.

8 High-Level Concept

The following slogan encapsulates ThreatLedger's mission and its distinctive approach to CTI sharing:

"***Secure Sharing. Real Benefits.***"

Here's why we chose it:

- **Security First**: "Secure Sharing" highlights the core feature of the platform: confidentiality, integrity, and control over sensitive data using blockchain technology.
- **Clear Incentives**: "Real Benefits" communicates that participation is not only secure but also rewarding. Users receive tangible advantages such as discounts, visibility, and premium access based on their contributions.
- **Trust & Simplicity**: The slogan is straightforward and communicates confidence. It encourages users to contribute without fear of data misuse or exposure, which is a major problem in many current CTI platforms.
- **Community & Impact**: It evokes a sense of collaborative value creation: by sharing intelligence, users improve the security posture of the whole network while earning personal rewards.

This high-level concept reflects the ambition of ThreatLedger to revolutionize CTI sharing through a decentralized ecosystem that improves privacy and is driven by rewards.

9 Key Metrics

We track four key metrics to measure the success of ThreatLedger:

(1) **Number of subscribers**: A direct indicator of market traction and recurring revenue. More subscribers means more revenue and validates the platform's value.

(2) **Number of active users**: How many users share or request CTI. It reflects user engagement and platform utility. This metric can be accurately measured by analyzing transactions recorded on the blockchain.

(3) **Average CTI quality score**: Derived from peer evaluations, it indicates the usefulness, relevance, and accuracy of shared threat intelligence.

(4) **Points usage rate**: Measures how earned points are redeemed for rewards, helping to assess how effective the incentive model is. Low usage for specific rewards may indicate that they are not attractive or relevant, helping decide whether to remove or adjust them.

Together, these metrics ensure that the technical infrastructure supports the community in building a sustainable, trusted, and high-quality CTI-sharing ecosystem.

10 Cost Structure

To deliver our product, we will require specific resources. Below, we present an estimated breakdown of the annual costs associated with these essential components (Italy market):

Development and Maintenance Costs:

- *Software Development and Salaries*: $\sim$ €120,000/year [7]
  The size of the software development team depends on the project timeline. If there is no urgency, a small team of three people (front-end developer, back-end developer, and project manager) may be sufficient.

- *Blockchain Integration and Security*: $\sim$ €50,000/year [6]
  This includes blockchain setup, maintenance, and related security. For this critical integration, we'll hire a specialized expert.

Operating Costs:

- *Server, Hosting, and Data Storage*: $\sim$ €10,000/year for server infrastructure and cloud storage.

Marketing and Customer Acquisition Costs:

- *Digital Marketing and Commercial Consulting*: $\sim$ €25,000/year [8]
  As a niche product, targeted advertising is sufficient. This cost includes consultation from a digital marketing expert or a sales consultant.

- *Partnership Development*: $\sim$ €10,000/year for travel, meetings, and partnership management.

Administrative Costs:

- *Office and Miscellaneous Expenses*: $\sim$ €40,000/year for office rent, utilities, and miscellaneous expenses in a medium-sized Italian city.

Estimated Total Annual Cost: $\sim$ €255,000.

11 Revenue Streams

Forecasting annual revenue is complex due to our tailored pricing, potential volume discounts, and various integrated incentive programs. These include platform points rewards that grant subscription discounts (detailed in Sec.5), alongside referral programs and tiered pricing structures. Below, we provide an estimation based on our different subscription models and service offerings.

- **Pro Plan Subscription**: Designed for small to medium organizations and security teams.
  - *Monthly Pricing*: €500 per user/month.
  - *Annual Pricing*: A discounted rate of €5,100 per user/year (approximately 15% off the monthly equivalent, €6,000/year).

  Revenue Hypotheses:

  - *Base Case (Mixed Adoption)*: Assuming 15 customers with an average of 3 users each on monthly plans, and 5 customers with an average of 3 users each on annual plans.

    $$(15\times 3\times €500\times 12)+(5\times 3\times €5,100) = €270,000+€76,500 = €346,500/\text{year}$$

  - *Optimistic Case (Higher Annual Adoption)*: Assuming 25 customers with an average of 4 users each on annual plans, and 5 customers with an average of 3 users each on monthly plans.

    $$(25\times 4\times €5,100)+(5\times 3\times €500\times 12) = €510,000+€90,000 = €600,000/\text{year}$$

- **Enterprise Plan Subscription**: Designed for large organizations requiring advanced features and dedicated support. Pricing is highly customized based on specific needs (e.g., user count, integration complexity). For estimation, we project securing 5 major customers in the first year, with an average annual contract value of €54,000 per customer (equivalent to €4,500/month).

  $$5 \times €4,500 \times 12 = €270,000/\text{year}$$

- **Premium Services and Customizations**: Additional services such as SIEM integrations, priority support, or specific custom development. Assuming a 30% uptake across all customers (estimated 25 total customers across Pro and Enterprise plans in the base case for illustrative purposes):

$$25 \times 0.3 \times €1,000 = €7,500/\text{year}$$

- **Strategic Partnerships and Integrations**: Potential revenue derived from collaborations with third-party vendors or governmental entities (e.g., revenue sharing agreements, licensing fees). These are not included in the current estimate due to the early stage of development.

- **Tailored Consulting and Support**: This stream covers specialized consulting, advanced training, or in-depth technical assistance beyond what is included in standard subscription plans (e.g., custom workshops, extended on-site support).

$$€20,000/\text{year}$$

Estimated Total Annual Revenue (Based on Pro Plan Base Case):

$$\begin{aligned} €346,500 \text{ (Pro Plan)} &+ €270,000 \text{ (Enterprise Plan)} \\ &+ €7,500 \text{ (Premium Services)} \\ &+ €20,000 \text{ (Tailored Consulting)} = €644,000 \end{aligned}$$

12 Conclusions

Imagining myself as an entrepreneur launching this company today, I believe that it has strong potential for success, while being aware of the challenges ahead. The cybersecurity market is rapidly growing, with the Italian sector reaching €2.48 billion in 2024 [5], driven by increasing cyberattacks and awareness of data protection. Modern threats, from APTs and ransomware to state-sponsored attacks, highlight the urgent need for advanced and collaborative defense mechanisms. At the same time, regulations such as GDPR demand solutions that ensure not only security but also privacy and traceability.

Our Cyber Threat Intelligence platform addresses these needs precisely, combining collaborative intelligence with blockchain technology to deliver security, transparency, and tangible incentives. The integration of Web 3.0 paradigms further strengthens its position as a modern decentralized solution.

For these reasons, if it met my needs, I would confidently invest in this company. Its value proposition is clear, concrete, and aligned with the current market demands. Personally, I would definitely use this solution: an indispensable tool to enhance corporate security posture and ensure compliance in an evolving digital landscape.

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